

Where the cubic-bound proof is right now

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Working fragment: classical regexes, in normal form, no bounded repetition $r^{\{n\}}$ (the “clean” fragment). Everything below is machine-checked in Isabelle with no gaps unless marked $[-\rightarrow]$ or $[o]$.

The goal (one inequality). Run the lexer’s normalized derivative automaton from r over input s , do *one* strongly-simplified step on the next letter c , open every resulting row into linear forms and deduplicate, and sum their sizes. That total is cubic in $|r|$:

$$\| \text{Open}(\Delta_c(\text{Front}(r, s))) \| \leq 2(|r| + 3)^3$$

uniformly in s (its length does not appear on the right).

Notation. $|r|$ = size of r ; $\text{acc}(r, k)$ = rows the derivative of r leaves in front of continuation k ; ∂k = one-step frontier of k ; σ = sequence normalizer; $w(r)$ = weight (letters, +1 per star) — always $w(r) \leq |r|$; $\text{Rows}(r)$ = all rows reachable from r (static); $\text{Front}(r, s)$ = the normalized front after reading s ; Δ_c = one strong simplify+prune step; Open = open rows to linear forms and dedup.

The chain (top to bottom). Read it as: each line is justified by the one below it.

[✓ **PROVEN**] **Goal, on the clean fragment.** $\|\text{Open}(\Delta_c(\text{Front}(r, s)))\|$ splits into a non-sequence part (already cubic) plus a sequence part bounded by the front total $\|\text{Front}(r, s)\|$. actual_union_gate_two_summands; rsize_set_rnonseq_members... _cubic

[✓ **PROVEN**] **Static front is cubic, for every input.**

$$\|\text{Front}(r, s)\| \leq (|r| + 3)^3 \quad \text{for all } s.$$

In words: however long the input, the normalized front never exceeds a fixed cubic in the original size. rsizes_afactored1_clean_static_cubic

[✓ **PROVEN**] **It rests on two facts about the static row universe:**

$$\#\text{Rows}(r) \leq |r| + 2 \quad \text{and} \quad \text{every row in } \text{Rows}(r) \text{ has size } \leq (|r| + 2)^2.$$

In words: few rows (linearly many), each small (quadratic). Linear $\times$ quadratic = cubic. card_apder_rows_clean_le_rsize_plus_2; apder_rows_member_size_quadratic

[✓ **PROVEN**] **The row count is linear because of the D law** (the hard part, cracked via the GPT-Pro “telescoping” invariant):

$$\#(\text{acc}(r, k) \setminus \partial k) \leq w(r) \quad (\text{clean } r, k).$$

In words: one derivative step adds at most $w(r)$ ($\leq |r|$) genuinely new rows. Proved by a single simultaneous induction on the stronger boundary invariant $\#((\partial(\sigma rk) \cup \text{acc}(r, k)) \setminus \partial k) \leq w(r)$ plus a strict-credit lemma. D_law_clean (from apder_T_bound/apder_S_bound via T_and_S)

[✓ **PROVEN**] **The D law applies to the *actual* rows.** The rows the gate feeds are in the clean domain. apder_clean_apder_rows; apder_clean_afactored1

$\rightarrow$ **CURRENT** **The one remaining step: glue the cubic front into the gate.** Push $\|\text{Front}(r, s)\| \leq (|r| + 3)^3$ through the single strong step, the opening, and the dedup

$$\text{Front}(r, s) \xrightarrow{\Delta_c} \cdot \xrightarrow{\text{Open}} \cdot \xrightarrow{\|\cdot\|} \leq 2(|r| + 3)^3,$$

and discharge the “clean” hypothesis for the actual root. *Known wrinkle:* the strong-step rows are only “tail-normal,” not fully clean, so the D law is used at the Rows/Front level and the strong opening is bounded on its own. Once this lands, the boxed goal is closed (clean fragment).

[◦ **LATER**] **After that:** lift the result from the clean fragment to *all* regexes including bounded repetition $r^{\{n\}}$. This needs a repetition-aware weight (the plain w fails when a $\{0\}$ block carries an empty-but-present frontier). Optional; may not be required for the thesis target.

The symbols, defined (so the inequalities above are self-contained). Throughout, r, k are regexes, Σrs a list-alternation, c a letter.

- **Size** $|r|$: $|0| = |1| = |c| = 1$, $|r_1 r_2| = 1 + |r_1| + |r_2|$, $|\Sigma rs| = 1 + \sum_i |r_i|$, $|r^*| = 1 + |r|$, $|r^{\{n\}}| = 1 + |r| + n$.
- **Frontier** ∂k — the rows of k after splitting its top alternation: $\partial 0 = \emptyset$, $\partial(\Sigma rs) = \bigcup_i \partial r_i$, $\partial k = \{k\}$ otherwise. (*A plain row is itself; an alternation splits into its branches.*)
- **Sequence normalizer** $\sigma(r, k)$ — “the row r followed by k ,” with units erased and right-associated: $\sigma(0, k) = 0$, $\sigma(1, k) = k$, $\sigma(r_1 r_2, k) = \sigma(r_1, \sigma(r_2, k))$; for an atom a (letter / star / alternation): $\sigma(a, 0) = 0$, $\sigma(a, 1) = a$, else $\sigma(a, k) = a k$.
- **Accumulator** $\text{acc}(r, k)$ — the rows that one derivative of r leaves in front of continuation k :

$$\text{acc}(0, k) = \text{acc}(1, k) = \emptyset, \quad \text{acc}(c, k) = \partial k, \quad \text{acc}(\Sigma rs, k) = \bigcup_i \text{acc}(r_i, k),$$

$$\text{acc}(r_1 r_2, k) = \text{acc}(r_1, \sigma(r_2, k)) \cup \text{acc}(r_2, k), \quad \text{acc}(r^*, k) = \text{acc}(r, \sigma(r^*, k)).$$

(*A letter hands its whole continuation-frontier forward; a sequence accumulates the left part against the normalized $r_2 k$ plus the right part against k ; a star re-enters itself. This recursion — one letter importing a whole frontier, balanced only across siblings — is exactly what made the D law hard.*)

- **Weight** $w(r)$ — letters, +1 per star; always $w(r) \leq |r|$: $w(0) = w(1) = 0$, $w(c) = 1$, $w(r_1 r_2) = w(\Sigma rs) = \text{sum}$, $w(r^*) = 1 + w(r)$, $w(r^{\{n\}}) = n(1 + w(r))$.
- **Static row universe** $\text{Rows}(r) = \{r\} \cup \partial r \cup \bigcup_q \partial q$, the union over all Antimirov sub-derivative terms q of r . (*Every row that can ever appear in front of r ; a finite set.*)

One-line status: **the hard combinatorics (the D law) and the static cubic front are done; one mechanical gluing step + a root-hypothesis discharge remain to close the gate on the clean fragment.** Full formal statement: CUBIC_OPEN_PROBLEM.pdf. Live detail: MAINLINE.md §2 and the PROGRESS_BACKREF.md tail.